

CSE4261: Neural Network and Deep Learning

Assignment-6

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Answer:

For the problem, we have used RestNet50 as a pretrained model and implemented GradCAM and IG for the original and adversarial image.

Visual Comparison



Figure 1: GradCAM: Original vs Adversarial Image

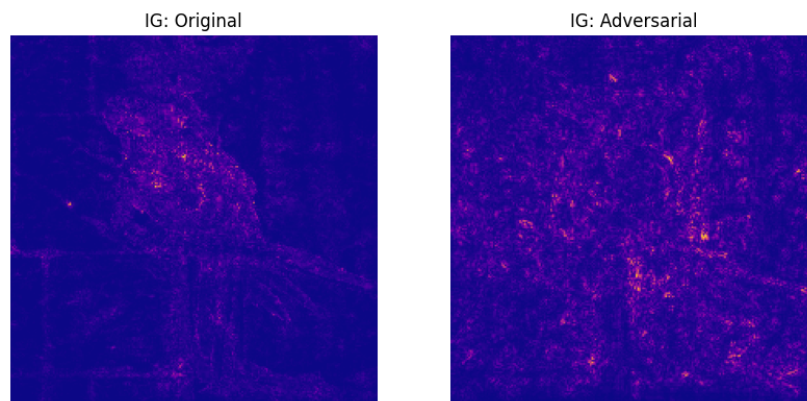


Figure 2: IG: Original vs Adversarial Image

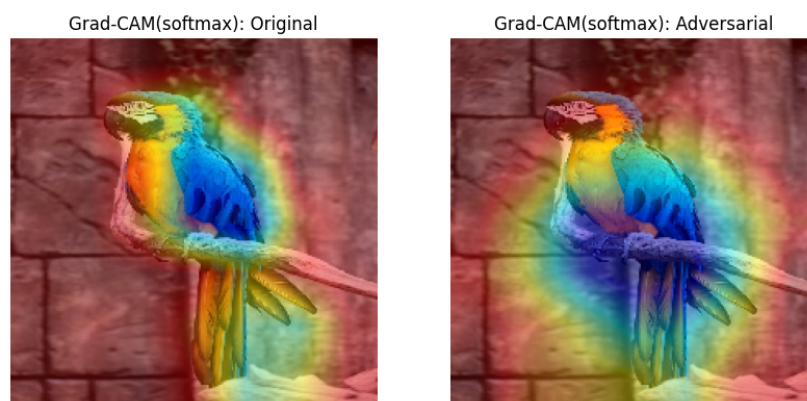


Figure 3: GradCAM: Using softmax layer

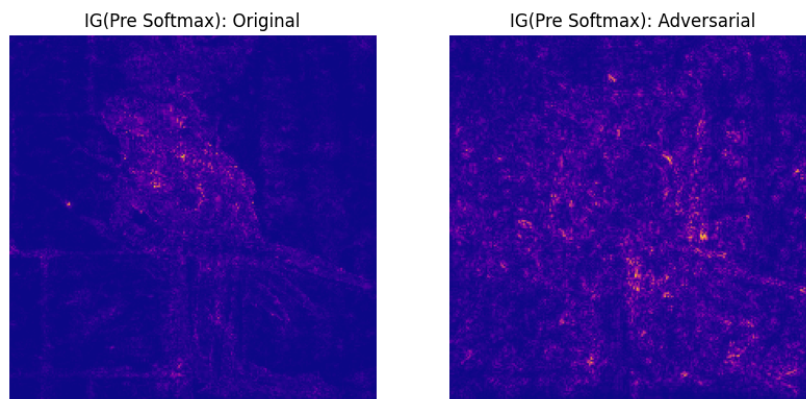


Figure 4: IG: Previous softmax layer

Discussion

In this assignment, we explored model interpretability techniques — **Grad-CAM** and **Integrated Gradients (IG)** — to analyze adversarial examples and understand which regions of an image influence the model’s predictions most significantly. For this, I have used ResNet50 as one of my favorite models trained with ImageNet.

1. Confusing Regions in Adversarial Examples

By applying Grad-CAM and IG to adversarially perturbed images, we observed that the neural network often focuses on *irrelevant or misleading regions* of the image. These regions are frequently *shifted* compared to the salient areas in the original image. For example, while the original image might highlight meaningful object features (e.g., a bird’s face), the adversarial image tends to redirect attention to the *background* or *edges*, which contributes to the model’s misclassification.

2. Comparison with Original Image

When comparing attribution maps of the *original image* and its *adversarial counterpart*, we noticed a significant difference. For the original image, both Grad-CAM and IG highlighted **semantically important regions**, aligning well with human perception. In contrast, the adversarial image’s heatmaps showed *dispersed or displaced activation*, indicating that the perturbation effectively manipulated gradient-based attention mechanisms, thereby **confusing the network**.

3. Effect of Using Softmax vs. Pre-Softmax Layers in Attribution Methods

According to the paper *“Pre or Post-Softmax Scores in Gradient-based Attribution Methods, What is Best?”*, using the softmax layer for Grad-CAM and the pre-softmax layer for Integrated Gradients (IG) is not recommended.

Softmax in Grad-CAM

Using softmax outputs in Grad-CAM can lead to **vanishing gradients** due to the saturation effect of softmax, especially when class probabilities are close to 0 or 1. This weakens the attribution maps and reduces their interpretability. Grad-CAM is more effective when gradients are taken concerning **pre-softmax (logit)** values.

Pre-Softmax in IG

IG is designed to work on output probabilities to remain **model-agnostic**. Using pre-softmax values in IG disrupts this principle and can lead to misleading results, since logits lack a direct probabilistic interpretation and may vary sharply in scale.

Grad-CAM should use **pre-softmax** scores, while IG should use **post-softmax** outputs. Reversing these choices can degrade explanation quality and cause unreliable interpretations.